

AMERICAN TERAWATT

HVDC Design

American Terawatt | July 22, 2026

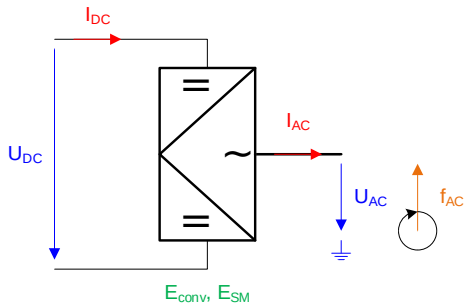
Control & Protection

Converter control and dynamic behaviour

What can a converter control?

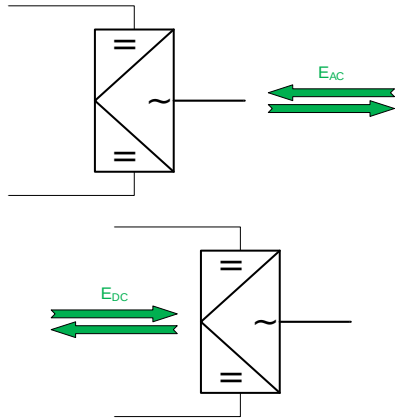
Control values of the fast control loop:

- DC voltage
- DC current
- AC current
 - Positive-phase active current $i_{p,psf}$
 - Positive-phase reactive current $i_{q,psf}$
 - Negative-phase active current $i_{p,nsf}$
 - Negative-phase reactive current $i_{q,nsf}$
- AC voltage
- AC frequency
- Converter energy



Energy flow

- The converter has only limited energy storage, so in steady state the converter energy must be balanced over one period.
- Energy balancing can be done through the AC side or through the DC side.
- One side defines the power flow, the other is responsible for the balancing.



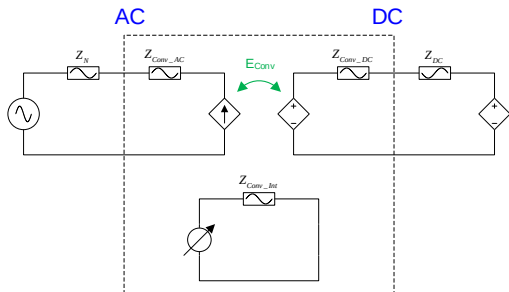
Basic control principle of the HVDC converter

Three major control loops:

- AC loop
- DC loop
- Inner loop

Impedances of the control loop:

- AC converter impedance
- DC converter impedance
- AC grid impedance
- DC line impedance
- Converter internal impedance



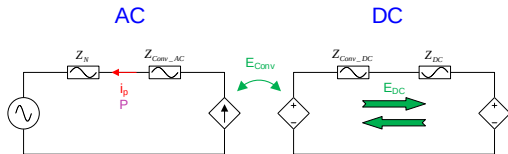
AC current control

Control input / reference: ω_{conv} , i_p / P

Control output: U_{dc}

Scenario

- AC power is defined by the operator.
- DC voltage controls the power flow on the DC side to balance the converter energy.



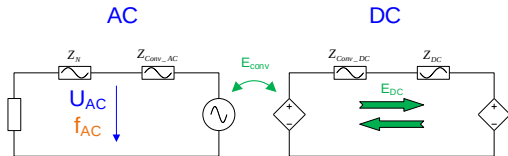
AC voltage control

Control input / reference: ω_{conv} , U_{AC} , f_{AC}

Control output: U_{dc}

Scenario

- AC power is defined by the AC load.
- DC voltage controls the power flow on the DC side to balance the converter energy.



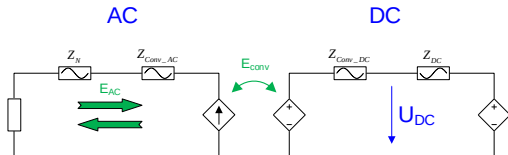
DC voltage control

Control input / reference: ω_{conv} , U_{DC}

Control output: i_p

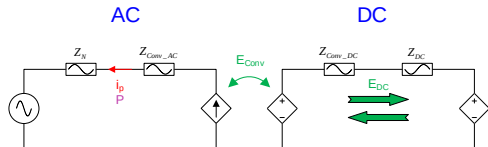
Scenario

- The station defines the voltage on the DC line.
- AC current (and therefore AC power) balances the converter energy.



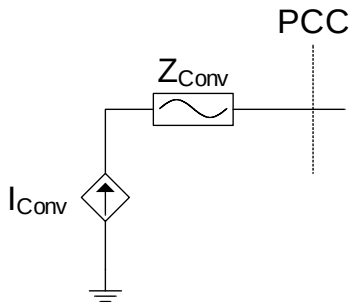
Reactive power control

- In control modes without AC voltage and frequency control, the reactive power can be controlled independently of the energy balancing.
- Reactive power does not require DC power flow and can operate without the DC line connected.

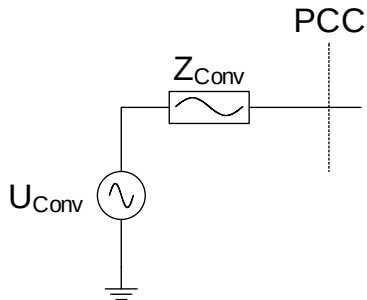


AC control - Current source vs. Voltage source

AC current source



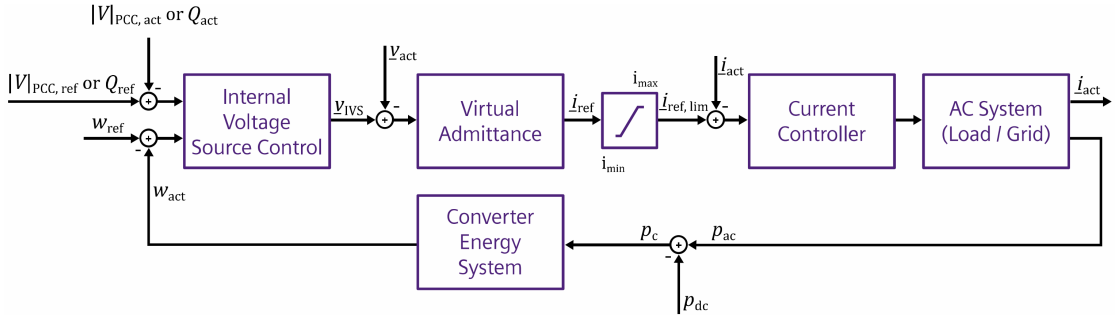
AC voltage source



AC control - Current source vs. Voltage source

	AC current source	AC voltage source
	Grid Following (GFL) P/Q control mode	V/f or Grid Forming (GFM)
Primary control parameters	Active current Reactive current	AC frequency AC voltage
Secondary control parameters	Active power Reactive power AC voltage	
Grid application	$SCL \geq 2$	$SCL < 2$

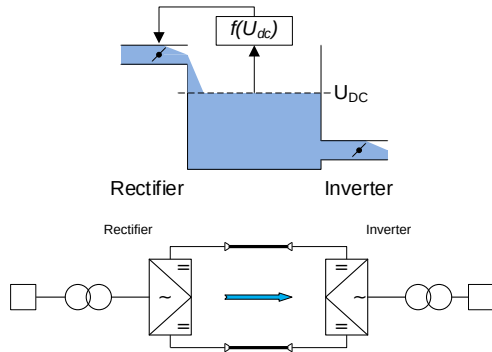
Grid-forming control concept



Grid-forming converters - advantages of grid-forming control in HVDC and FACTS applications (white paper, Siemens Energy).

Steady-state power transmission

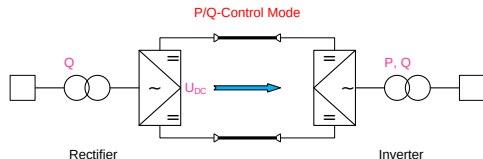
- One station controls DC voltage.
- One station controls active power.
- Two active-power control modes are possible:
 - P/Q control mode
 - U/f control mode



Steady-state control principle

Standard operation

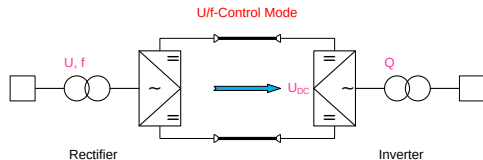
- The station that sees the highest DC voltage controls the DC voltage.
- Active power can be defined at any station.



Steady-state control principle

Wind-farm connection

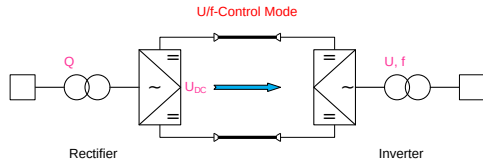
- Rectifier receives power coming from the WTG.
- Inverter must consider the DC line voltage drop.



Steady-state control principle

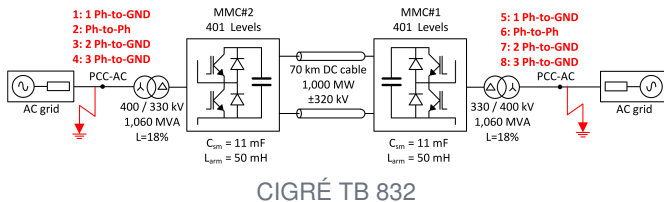
Power supply to islanded network

- Inverter provides power required by the loads
- Rectifier controls DC voltage by providing the required active power to the DC side



Dynamic analysis

- Dynamic analysis covers converter behaviour during AC grid faults (on the network side of the converter transformer).
- AC grid faults are very common, so the converter must stay connected and provide supporting functions.



Dynamic analysis: 3ph-to-GND at the inverter

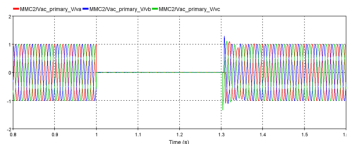


Figure 5-6 (a) - MMC2 - Grid-side voltages

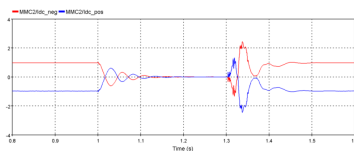


Figure 5-6 (d) - MMC2- DC-side currents

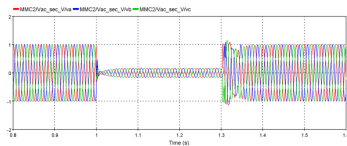


Figure 5-6 (b) - MMC2 - Converter-side voltages

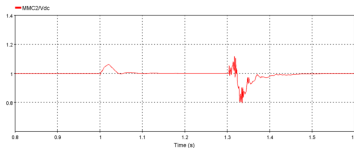


Figure 5-6 (e) - MMC2- DC voltage

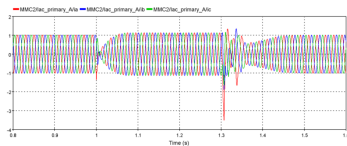


Figure 5-6 (c) - Grid-side currents

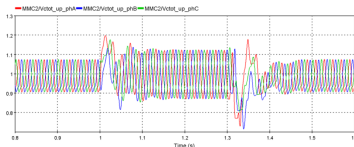


Figure 5-6 (f) - MMC2 - Sum of capacitor voltages (upper arm)

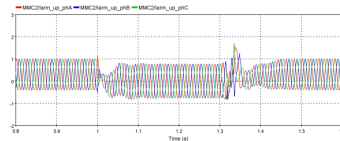


Figure 5-6 (g) - MMC2 - Arm currents in upper arm

Figure 5-6 3ph-ground fault at the inverter station typical waveforms

Dynamic analysis: 3ph-to-GND at the rectifier

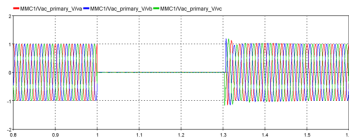


Figure 5-7 (a) - MMC1- Grid-side voltages

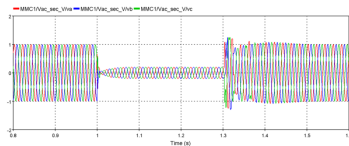


Figure 5-7 (b) - MMC1- Converter-side voltages

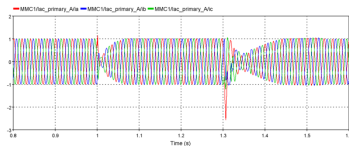


Figure 5-7 (c) - Grid-side currents

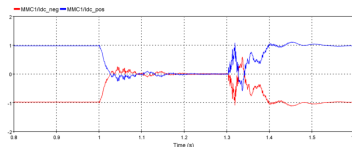


Figure 5-7 (d) - MMC1- DC-side currents

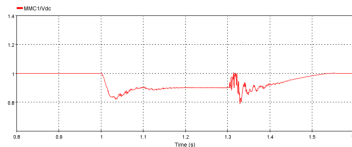


Figure 5-7 (e) - MMC1- DC voltage

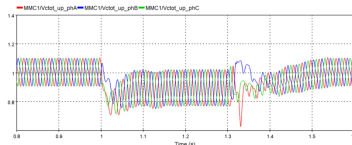


Figure 5-7 (f) - MMC1- Sum of capacitor voltages (upper arm)

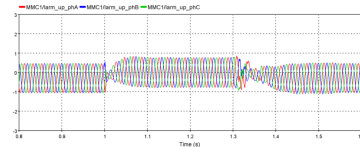
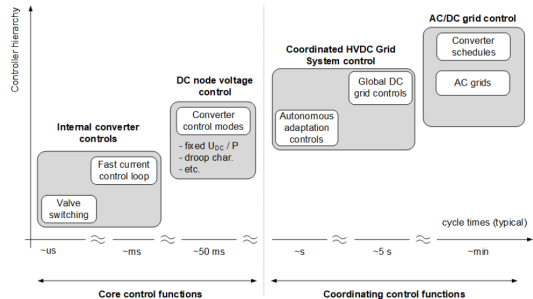
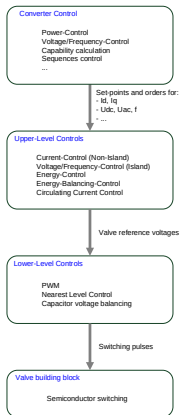


Figure 5-7 (g) - MMC1 - Arm currents in upper arm

Figure 5-7 3ph-ground fault at the rectifier station typical waveforms

CIGRÉ TB 832.

Control hierarchy



CLC/TS 50654-1:2020 - HVDC control

Control requirements

Input parameters for the control:

- Voltage measurement
- Current measurement

Control functions:

- Current control
- Voltage control
- Reference voltage generation for six voltage sources
- Voltage synthesis by generating switching pulses for individual modules

